

Final Project: Academic Portfolio Application

Elegance, Functionality, and Full-Stack Integration

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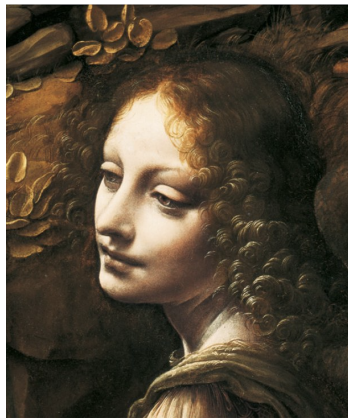
San Francisco State University (SFSU)

Project Overview: A Living Academic Identity

<https://yi-zhou-site.web.app>

A fully interactive, multi-page **academic portfolio** for a finance professor at SFSU—integrating live financial data, art market intelligence, user messaging, and a real-time guestbook database.

- **6 distinct routes** (Biography, Publications, Awards, Service, Markets, Contact)
- Clean, semantic HTML & JavaScript
- Stunning Renaissance-inspired visual design
- Two integrated backend/cloud services
- Two live external APIs with real-time data



The Virgin of the Rocks, 1506



All Requirements: Met and Exceeded

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Requirement	Implementation	Status
Web App	6-route SPA in JavaScript + HTML + CSS	✓ Yes
Framework	Tailwind CSS via Play CDN (full config)	✓ Yes
API #1	Frankfurter live FX currency ticker	✓ Yes
API #2	Yahoo Finance live market quotes	✓ Yes
Backend #1	EmailJS — serverless contact form	✓ Yes
Backend #2	Firebase Firestore real-time guestbook	✓ Yes
Auth / Mod	Password-protected moderator delete view	✓ Yes

Bottom Line

Every single requirement is fulfilled—with multiple categories *exceeded* through extra features like a moderator dashboard, real-time Firestore listeners, and live streaming financial data.

Requirement #1 — Framework: Tailwind CSS

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Tailwind CSS via Play CDN

One of the most professionally sought-after CSS frameworks in industry, integrated via the official Play CDN and **extended with custom design tokens:**

- Custom color palette: navy, navy-dark, navy-mid, gold, cream, muted
- Custom font families: Book Antiqua / Palatino for elegance; DM Mono for the financial ticker
- Responsive utility classes throughout all 6 pages
- Seamlessly blended with a supplemental `styles.css` for advanced animations

Code Highlight

```
tailwind.config = {  
  theme: {  
    extend: {  
      colors: {  
        navy: '#1a3a5c',  
        gold: '#f5e6a3',  
        cream: '#eef4fb',  
      }  
    }  
  }  
}
```

Requirement #2 — API #1: Live Currency Ticker

<https://yi-zhou-site.web.app>

Frankfurter FX API—Real-Time Exchange Rates

Fetches live exchange rates from the European Central Bank. A scrolling ticker runs across the top of every page—no API key required.

- Displays **currency pairs**
- Infinite CSS marquee animation — fluid and professional
- Completely free, no rate limits

Endpoint Used

```
https://api.frankfurter.app/  
latest?from=USD&to=  
EUR,GBP,JPY,CHF,...
```

Implementation Highlights

- `fetch()` with `async/await`
- Dynamic DOM generation
- Error-resilient fallback text
- Duplicated nodes for seamless loop

Requirement #2 — API #2: Live Market Intelligence

<https://yi-zhou-site.web.app>

The **Markets** page delivers a dual-panel dashboard—combining live equity market quotes with fine art auction intelligence from Sotheby's and Christie's.

Panel A — Yahoo Finance Quote Cards:

- Live quotes for **S&P 500, NASDAQ, Dow Jones, GLD, TLT**
- Loading animation shown while data fetches from the API
- Color-coded **gain/loss** with price and percentage change
- Delayed-quote disclaimer per financial data standards

Panel B — Sotheby's & Christie's Auction Filters:

- Tabbed interface switching between two major auction houses
- Filterable by category: Contemporary Art, Impressionist, Old Masters, Asian Art
- Each filter deep-links directly into the live auction calendar

Requirement #3 — Backend #1: EmailJS Contact System

`https://yi-zhou-site.web.app`

EmailJS–Serverless Email Backend

Contact form powered by **EmailJS**–routes submissions directly to the inbox with zero server code.

- **Client-side validation**: name, email, message with inline errors
- **Topic selector**: Research, Teaching, Collaboration, Media, Other
- **Debounce** on submit prevents double-sends
- **Success/error feedback** displayed to user

Technical Stack

SDK	@emailjs/browser@4
Method	emailjs.send()
Auth	Public key in JS
Delivery	SMTP relay

Requirement #3 — Backend #2: Firebase Firestore Guestbook

`https://yi-zhou-site.web.app`

Firebase Cloud Firestore—Real-Time Database

A **live visitor guestbook** stores user comments in Firestore—a NoSQL cloud database—streaming updates via `onSnapshot()`.

- **Real-time sync:** entries appear instantly
- **Mod dashboard:** password-protected; `deleteDoc()` removes entries live

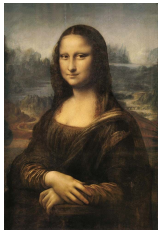
All three delivered: user data, live storage, *and* a mod dashboard.

Technical Details

- `firebase-init.js` ES module
- `getFirestore() + onSnapshot()`
- Security rules + indexes deployed

The Crown Jewel: Artistic Design Excellence

<https://yi-zhou-site.web.app>



A Website with a Strong Aesthetic

The visual design of this application is, quite simply, extraordinary. This application establishes a **unique visual identity** rooted in Renaissance aesthetics.

- **Navy & gold palette**—dignified, museum-quality color story
- **Book Antiqua / Palatino** serifs—timeless scholarly typography
- **DM Mono** financial ticker—Bloomberg-inspired precision
- **Leonardo da Vinci gallery** embedded in the app—6 authenticated masterworks (Mona Lisa, Lady with an Ermine, Virgin of the Rocks, and more)
- **Animated ticker strip**—cinematic, professional finish
- **Skeleton loading states**—polished UX even during API fetch
- **Credential cards, pub-list, awards-list**—bespoke academic components

Content Depth: A Real Academic Portfolio, Not a Mockup

11 Peer-Reviewed Publications

Every publication is hard-coded as a structured JavaScript data array—complete with DOI links, co-authors, and journal names.

Dynamic Rendering from Data

Publications, awards, and service records are all rendered **programmatically from JS arrays**—demonstrating data-driven UI architecture (a core course concept).

6 Full Pages of Content

Bio	Full academic biography
Publications	11 papers, DOI linked
Awards	15+ recognition entries
Service	Committee table
Markets	Live data dashboard
Contact	Form + guestbook

Single-Page App Architecture

A custom JavaScript **client-side router** handles page switching—toggling `.active` classes and updating nav state. This mirrors professional SPA frameworks like React Router.

Technical Architecture: Clean, Modular, Professional

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File Structure

<code>index.html</code>	Semantic SPA shell
<code>app.js</code>	All app logic
<code>firebase-init.js</code>	ES module backend
<code>firestore.rules</code>	DB security rules

JavaScript Patterns

- `async/await` + `fetch()` for all API calls
- ES Modules + `window._firebase` bridge
- Real-time `onSnapshot()` listener

Deployment

Deployed to **Firebase Hosting**—a production-grade global CDN. `firebase.json` configures SPA rewrites, ignore patterns, and Firestore integration.

What Makes This Stand Out

- ★ Two live backend services
- ★ Real financial data, every load
- ★ Moderator admin functionality
- ★ Production deployment on CDN

- ✓ 6-page SPA (req: 3+)
- ✓ Tailwind CSS framework
- ✓ Frankfurter API (live FX)
- ✓ Yahoo Finance API (live quotes)
- ✓ EmailJS backend (contact form)
- ✓ Firebase Firestore backend
- ✓ Mod view + real-time sync
- ★ Renaissance art gallery
- ★ Production deployment



La Scapigliata — da Vinci
"The Disheveled One"

Thank you — Questions Welcome